Chapter 4 : Model Questions

Q1.	Write Python statements using NumPy and Matplotlib to create a Figure and an Axes object, and plot the function ($y=x^2$). Use different background colours for the Figure and Axes regions so that the two objects can be visually distinguished.
Q2.	What is the purpose of a **legend** in a Matplotlib graph? Explain its significance when multiple curves are plotted on the same Axes object. Write a Python program using **NumPy and Matplotlib** to plot the following two functions on the same Axes object: $y1=x^2$ $y2=3x$ The program should satisfy the following requirements: 1. Label the **x-axis** as x and the **y-axis** as y. 2. Add appropriate labels to both curves and display a **legend** to identify them. 3. Set the **line width of the first curve** to **3** and its colour to **red**. 4. Set the **line width of the second curve** to **2** and its colour to **green**. 5. Display the resulting graph with an appropriate title.
Q3.	**Consider the following functions:** $y1=\sin(x)$ $y3=\sin(3x)/3$ $y5=\sin(5x)/5$ $y7=\sin(7x)/7$ Write a Python program to: 1. Generate 500 equally spaced values between 0 and 6.3. 2. Plot all functions on the same axes. 3. Use four different line styles. 4. Plot the sum using a thicker black line. 5. Add x-axis, y-axis and grid.
Q4.	Write a Python program using Matplotlib to plot the function $y=x^2-4$ and modify the position of the **x-axis and y-axis** so that they intersect at the origin (0,0). Hide the top and right spines of the plot.
Q5.	Write a Python program using Matplotlib to plot the function $y=x^2$ for x values from -5 to 5. Use the annotate() method to mark and label the point (2,4) on the curve.
Q6.	Write a Python program using Matplotlib to create a Figure and manually position an **Axes object** within the Figure using the add_axes() method. Plot the function y=x2 for $0 \le x \le 5$. The program should: 1. Create a Figure object. 2. Create an Axes object using fig.add_axes([left, bottom, width, height]). 3. Use suitable values for left, bottom, width, and height to position the Axes within the Figure. 4. Plot $y=x^2$in the Axes. 5. Label the x-axis and y-axis. 6. Give the graph an appropriate title. 7. Display the Figure.
Q7.	Write a Python program using Matplotlib to create a Figure containing **four subplots arranged in a 2 × 2 grid** using the add_subplot() method. Plot the following functions in the four subplots:$y=x$, $y=x^2$, $y=x^3$, $y=\sqrt{x}$
Q8.	Explain the purpose of the add_subplot() method in Matplotlib. Write Python statements to create a Figure containing **three vertically arranged subplots** using add_subplot(), and plot $y=x$, $y=x^2$, and $y=x^3$ in the three subplots.
Q9.	Consider the following Python statements:

	fig, ax = plt.subplots(2, 2) 1. How many Axes objects are created? 2. What is the shape of the ax object? 3. How would you access the Axes object in the first row and second column? 4. How would you access the Axes object in the second row and first column? 5.Write statements to plot ($y=x^2$) in the first subplot and ($y=x^3$) in the last subplot.
Q10.	Develop a Python program using plt.subplots() to create **four Axes objects in a 2 × 2 arrangement** and plot the following: • First Axes: ($y=x^2$) • Second Axes: (y=2x+1) • Third Axes: (y=sin(x)) • Fourth Axes: (y=cos(x)) The program should include: • Appropriate titles • x- and y-axis labels • Grid lines • Different line colours • tight_layout() • Display of the final Figure
Q11.	Differentiate between the subplots() function and the add_subplot() method in Matplotlib. Explain their purpose and syntax with suitable examples. Also write Python statements to create **four subplots arranged in a 2 × 2 grid** using: 1. plt.subplots() 2. fig.add_subplot()
Q12.	Write a Python program using Matplotlib to demonstrate how an Axes object can be inserted inside another Axes object. Create a main Axes to plot the function y=sin(x), $0 \leq x \leq 10$ and create a smaller inset Axes inside the main Axes using the add_axes() method. show only the portion from x=2 to x=4. Label the main plot appropriately and give suitable titles to both the main and inset plots.
Q13.	Write a Python program using **NumPy and Matplotlib** to generate a **noisy sinusoidal signal** and display it using a main Axes object. Create a second, smaller **inset Axes object** inside the main plot and use it to show an **enlarged view of a selected portion** of the noisy sinusoidal signal. The program should: 1. Generate a sinusoidal signal with random noise. 2. Create a Figure and a main Axes object using add_axes(). 3. Plot the complete noisy signal in the main Axes. 4. Create a smaller Axes object inside the main Axes. 5. Plot the same signal in the inset Axes. 6. Set suitable axis limits in the inset Axes to display an enlarged portion of the original signal. 7. Add suitable labels and titles.
Q14.	Write a Python program using **Matplotlib** to plot the function y=x^2 for (x)values ranging from **1 to 10,000**. Display the graph using a **logarithmic scale on the x-axis**. The program should: 1. Create a Figure and Axes object. 2. Generate suitable values of (x) from 1 to 10,000. 3. Calculate (y=x^2). 4. Plot the function. 5. Set the x-axis to a **logarithmic scale** using set_xscale(). 6. Label the x-axis and y-axis. 7. Give the graph an appropriate title. 8. Display the graph.

Q15.	Write a Python program using NumPy and Matplotlib to plot the functions $y1=(x-3)^2$ and $y2=-(x-2)^2+8$ for ($0 \leq x \leq 5$). Use the fill_between() method to colour the region between the two curves where ($y2 \geq y1$). Label both axes and give the graph an appropriate title.